

A Flow-Based Centralized Route Guidance System for Traffic Congestion Mitigation

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Abstract—We propose a new route guidance system (RGS) to mitigate traffic congestion called Flow-based Capacity-aware Rerouting (FCR), which achieves better global optimality by considering the capacity of detour routes. FCR introduces a flow-based traffic management method that enables us to compute detour paths in units of flow rather than per vehicle processing, which reduces the load to calculate detour paths and makes the centralized system practically feasible. With the flow-based traffic management, FCR also enables us to consider traffic volume that should bypass the congested roads to mitigate the congestion and suggest detour paths to the corresponding traffic volume of vehicles. By prioritizing travel time loss when choosing detour routes to apply, FCR leads to better global optimality with the limited volume of rerouting traffic while considering each rerouted vehicle's benefit, such as travel time loss. FCR offers a new centralized system architecture for actual real-time traffic control, which improves both global optimality and each vehicle's benefit within a feasible computational load.

Index Terms—Intelligent transportation system, vehicular congestion avoidance, route guidance system.

I. INTRODUCTION

RAPID growth in the number of vehicles is causing heavy traffic congestion in cities worldwide. Traffic congestion takes up people's time and causes significant economic loss [1], [2], [3], and it also has a severe negative impact on the environment [4], [5]. As automobiles are becoming increasingly automated and connected in recent years, it is expected to alleviate traffic congestion using dynamic detour routes. The vehicle rerouting strategy in the face of traffic congestion in urban areas has gained increased interest due to the development of intelligent transportation systems (ITS) [6]. Currently, a method called route guidance system (RGS) is attracting attention to reduce losses due to traffic congestion [7]. In the RGS studies, it is assumed that traffic conditions can be monitored by road side units (RSU) or via vehicle connections. When traffic congestion occurs, optimal routes according to the traffic state are computed and suggested to the drivers to improve traffic efficiency and mitigate that congestion.

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Since minor traffic congestion on a single road segment may grow into heavy ones [8], avoiding the loaded road segments by rerouting is adequate to mitigate congestion [9]. Several rerouting schemes have been proposed to alleviate traffic congestion when congestion on some road segment is detected [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. One typical architecture for such systems is the centralized system, which typically collects traffic condition measurements for each road segment or intersection on a central server, calculates the optimal routes, and suggests the routes to encourage vehicles to choose better routes. Although this centralized strategy is good in terms of global optimization, it has major drawbacks that its heavy computational load makes it hard to put it to practical use. In addition, the global decision is not always good from the viewpoint of the benefit of individual vehicles, e.g., sometimes the suggested detour routes may result in longer travel time than not rerouting, and this may cause an unfair feeling of users.

Several studies then proposed distributed approaches to achieve feasible systems in terms of computational load [23], [24], [25], [26], [27], [28], [29], [30]. The distributed approach would also be good in terms of benefits to individual vehicles since each vehicle can choose routes to maximize its own benefits. Typically in their studies, the infrastructure collects and shares traffic conditions with distributed agents such as RSUs. The agents or vehicles themselves compute detour routes to reduce the travel time of vehicles reaching their destinations. However, since the information shared in each agent is inherently limited, and also since vehicles' new routes easily conflict with one another, it results in suboptimal solutions and often leads to secondary congestions on the detour routes of vehicles, as is shown in Sec. IV-F of this paper.

In this paper, we propose a new centralized scheme that overcomes the shortcomings of both centralized and distributed approaches in the literature. Although our method is designed in a fully centralized manner, we newly introduce flow-based traffic amount measurements that significantly reduce the cost of computing rerouting paths. We define *flow* as the stream of vehicles that travels between two major intersections. By managing the entire traffic as a collection of flows, and computing detour routes in the units of flows rather than vehicles, we can save computational load significantly. The flow management also enables us to estimate the required volume of rerouting vehicles. When congestion is detected, we calculate the traffic volume needed to avoid the congested road to mitigate the congestion and reroute that traffic volume

to bypass the congested road considering the capacity of the detour routes. This flow-based approach enables us to make high-performance optimization worthy of the centralized approach, with a lower feasible computational load.

We propose a new rerouting paths computation method called flow-based capacity-aware rerouting (FCR), designed based on the flow-based traffic management scheme to reduce the computational cost. Simultaneously, however, it also considers the benefit of individual vehicles, such as the time delay experienced by the drivers of vehicles, which is a merit of the distributed approach. Our idea is to exploit the following fact: Although most of the centralized methods in the literature provide rerouting paths only to vehicles traveling near the congested road, vehicles traveling relatively far from the congested road will have detour paths with less travel time loss in a high probability. That is, by handling a larger number of detour route candidates that can be chosen by vehicles traveling far from the congested roads, and by selecting the detour routes with the priority of traveling time loss, we can achieve traffic control that improves not only global optimality but also individual benefit. By taking the benefits of individual vehicles into account, people would be motivated more to follow the suggested routes, which would further improve the global optimization performance. Satisfying both the global optimization and the individual benefits will be an ideal property for RGS.

The main contribution of this paper is summarized as follows.

1. We propose a flow-based rerouting path computation scheme that reduces the computational cost and makes the centralized approach feasible to cover a large roadmap area.
2. By selecting less expensive alternative routes chosen by vehicles running farther away from the congested road segments, we take the benefit of the rerouted vehicles into account and make users motivated to follow the route suggestion.
3. We achieve better global optimization than the conventional centralized methods by considering the capacities of road segments in the detour routes and by prioritizing the vehicles' route candidates based on travel time loss.

Note that this paper does not compare the performance of the centralized approach with the hybrid or distributed ones, which are described in Sec. II. We provide techniques that augment the capability of the centralized approach and demonstrate their feasibility in terms of computational load and optimization performance. This paper is an extended version of [31], from which we updated the entire article description. Specifically, we mainly (1) updated the claim of contribution, (2) extended the road capacity model to consider the capacity of intersections, (3) extended the methodology to cover simultaneous multiple congested roads, and (4) provided more extensive evaluation results.

The remainder of this paper is organized as follows. In Section II, we describe the outline of RGS, its related work, and its problems. In Section III, we describe the proposed method. In Section IV, we evaluate the proposed method with the traffic flow simulator SUMO, and finally, in Section VI, we conclude our study.

II. RELATED WORK

A. Detour Route Computation

There is a large body of literature on the optimization of urban transportation. Two major means for optimization are traffic signal control and route control. The latter is particularly valuable to mitigate congestion once it occurs, and thus many contributions exist that detect and mitigate congestion by rerouting a part of vehicles [32].

The most basic detour path computation algorithm is called Dynamic Shortest Path (DSP) [10]. The shortest paths are used to reroute vehicles under some dynamic metrics such as travel time, travel distance, road density, etc. Rezaei et al. reroute vehicles to the minimum travel time (TT) path based on its congestion level estimation [22]. Yamashita et al. also use the minimum TT path based on their own flow analysis [12]. Wang et al. proposed a quick shortest-path computation method based on G-Tree to catch up with the traffic dynamics [13].

One of the problems in the shortest-path strategy is the traffic concentration in the detour paths since every detour path is usually computed from the same weight function. To avoid the problem, we should consider cooperative path computation, balancing the traffic load among roads widely on the map. Souza et al. update the weight of links every time it computes the shortest paths for a vehicle to distribute traffic according to traffic density [17]. Wang et al. tried to distribute traffic by following real-time congestion levels and avoiding rerouting to the congested roads [13]. However, the weight-based cooperative approach results in rough load balancing and is less effective.

Another effective means to distribute traffic is to use multiple detour paths. Pan et al. demonstrated that distributing traffic into the k -shortest paths [33] is effective in load balancing [11]. They proposed several k -shortest paths usages such as Random k -Shortest Path (RkSP), Entropy Balanced k -Shortest Path (EBkSP), Flow Balanced k -Shortest Path (FBkSP). Souza et al. distributes traffic among k -shortest paths according to Boltzman distribution [14]. However, since those methods basically distribute traffic randomly or according to some simple statistic, traffic tends to concentrate on particular roads in the detour paths, especially in case multiple congested roads appear. Note that multiple or area congestion easily occurs since congestion tends to be extended quickly [8].

Further studies for cooperated routing have been conducted to improve the optimality of the rerouting schedule. Pan et al. proposed to consider the footprint of vehicles [11]. They proposed a heuristic shortest paths computation algorithm called A^*R that extends the basic A^* algorithm to penalize repulsion paths that are inconvenient as rerouting paths. Liu et al. proposed a method of balancing traffic load among roads by coordinating between routes used by vehicles through bidding [29]. Guidoni et al. in [26] and Liu et al. in [20] consider each vehicle's traveling paths in real time and try to reflect them on the metric of the weight function of path computation. However, these methods still lack the consideration of global optimization based on road capacity, and the optimization performance is still suboptimal.

Our method, FCR, takes the capacity of detour paths into account and aims to achieve global optimization by prioritizing detour paths in terms of travel time loss. To the best of our knowledge, no work is proposing a capacity-based optimization of rerouting paths in the literature.

B. Architecture: Centralized Vs. Decentralized

There are several types of RGS system architecture in the literature, including centralized and decentralized configurations. It is essential to see the difference in characteristics between centralized and decentralized architectures in the RGS system. Here, we classify the architecture into centralized, decentralized, and hybrid to see their pros and cons within the literature.

We first see the centralized approach. Pan et al. proposed a real-time RGS system that monitors and controls traffic to avoid congestion [11]. In their system architecture, the central server (1) collects traffic measurements, (2) predicts collision from the measurements, (3) selects vehicles to reroute, and (4) pushes reroute guidance to the vehicles. Communication infrastructure is needed to collect data and provide guidance to the vehicles, such as cell phone communication (as is done in Google Maps [34] or TomTom [35]) or RSU [36]. Since the centralized system can be designed to use complete information collected from the target area, global optimization is possible, and the optimality is generally good.

However, the weakness of the centralized approach is the computational load to control the whole area of a city [22], [27]. Several static optimization methods using techniques such as simulated annealing [15] or mixed integer and linear programming (MILP) [16] exist. Still, they are impractical since they need significantly large computational power. Thus, heuristic algorithms based on the (single or k -) shortest paths computation have been a major strategy in this study area. Nevertheless, the computational load is not feasible in practice because the central server usually computes detour routes per vehicle in a dynamic transition of the traffic state. A methodology that combines high optimality with low computational load is required.

Another issue in the centralized approach is the fairness among vehicles [21]. Rerouting paths computed for global optimization are often unfair and sometimes rather not beneficial for a part of vehicles, i.e., it takes a larger time if they follow the reroute guidance. In this case, some vehicles do not follow the reroute suggestion of RGS, and the optimization performance degrades. One of the compromised solutions is to ensure benefits such as travel time reduction when following the suggested detour paths. Even if it is not strictly fair among vehicles, this solution will work in practice. Note that the current solutions for the fairness problem are all distributed approaches in which local agents such as vehicles or RSUs compute and select routes for their own benefit [23], [24].

Hybrid approaches that combine central control and distributed reroute computation could be a choice to solve both computational load and fairness problems. It usually collects traffic measurements using infrastructure (such as RSUs and cell phone networks), shares them among local agents such

as vehicles, RSUs, or some virtual entities, and the local agents compute routes to apply. In [22], RSUs work together to detect traffic congestion, inform vehicles of road states, and vehicles compute their own detour paths that minimize their travel time. In [24], the infrastructure monitors the traffic state via GPS-based location measurements, detects congestion, and sends the congestion information to vehicles to support each vehicle's rerouting decision. These methods avoid the fairness problem by making vehicles responsible for their routes, but their selfish route selection may cause secondary congestion in the detour paths. In [25], Qi et al. proposed a method to deliver information on traffic incidents that causes congestion to selected drivers to help them detour in urban areas, and showed the effect of information limitation.

As is already mentioned, cooperative path selection has been studied to solve the problem. Liu et al. proposed a bidding-based dynamic route planning method in which routing decision is made cooperatively through negotiation among vehicles [29]. In [26], the centralized system measures the density of roads, and vehicles update the density of routes when vehicles decide to reroute, which avoids traffic concentration on certain roads. Pan et al. provide a decentralized implementation of EBkSP and A*R in which footprints of vehicles are used to distribute traffic [27]. These cooperative approaches improve traffic load balance and optimize traffic conditions. However, there is a problem of communication load to share the traffic conditions of the entire map with each vehicle, which naturally leads to suboptimal traffic control.

On the other hand, several fully distributed architectures have been proposed. Typically, each vehicle monitors the road state to detect congestion and shares the information among vehicles via direct or multi-hop V2V communications without any infrastructure. Since there is no infrastructure support, the primary challenge in this approach is to collect road states [18]. To facilitate efficient collection of traffic state, Khan et al. in [28] and Lin et al. in [23] proposed to create clusters of vehicles, and the cluster heads are responsible for communicating with other clusters. Akabane et al. apply the approach of social networks to vehicular networks to rank vehicles to measure traffic states, and vehicles collaboratively compute cooperated rerouting paths to avoid congested spots [30]. However, the bottleneck of this method is the inability to collect traffic conditions from an entire large area.

It is worth noting that next road rerouting (NRR) [21] is a centralized approach but does not suffer from either the computational load or the fairness problem. NRR assumes the SCATS [37] system, which has already been in use in more than 27 countries. When NRR detects a congested road, it guides the vehicle around it by suggesting only the direction, i.e., the next road, to go from the intersection instead of showing an entire route to the destination. This system actually works to mitigate congestion, but the ability is limited since next-hop roads often cannot indicate an effective detour route.

In recent years, several machine learning-based techniques have been proposed to control vehicular traffic. Machine learning techniques are useful because they can predict future traffic flows and congestion [38], [49], [50]. Currently, however, machine learning only provides predictive capabilities. It is not

yet possible to create an optimized traffic schedule to alleviate congestion after it has occurred.

III. PROPOSED METHOD

A. Overview

Our method, FCR, works based on flow-based measurements of traffic conditions. We assume that traffic conditions on each road in the target area are collected on a central server in the cloud via infrastructures such as RSUs or cellular communications. It is also assumed that the destinations of all vehicles in the target area are collected on the server. Furthermore, we assume that the planned routes of all vehicles are collected, but the routes can actually be estimated by the shortest route calculation. As a result, the traffic volume moving between each intersection pair, i.e., from any origin to destination intersections, can be monitored. We define the traffic volume between two intersections as the *flow volume*, and FCR monitors the flow volume between every major pair of intersections. By capturing flow volume, it is possible to determine the traffic volume that can be diverted from each intersection to alleviate congestion.

When congestion is detected from the monitoring information, FCR determines the volume that should be diverted from the congested road segment. FCR then computes the rerouting paths and builds a rerouting table that holds information about the detour paths, such as the intersection pairs (i.e., the origin and destination of detour paths), the capacity of the paths, and the estimated travel time loss (i.e., the difference in travel time between the shortest and the detour paths). In case of congestion in multiple road segments, FCR builds multiple rerouting tables for each congested road segment. FCR selects a set of detour paths from the rerouting tables that satisfies the total traffic volume required to detour, prioritizing travel time loss. The selected set of detour paths is applied in route guidance to reduce the required amount of traffic entering the congested road segment by a specified volume. Route guidance is provided in a stochastic manner on the road before entering the detour route. Traffic is assumed to be properly guided using infrastructure such as RSUs and cellular communications.

We illustrate the idea of FCR in Fig. 1. In this figure, we focus on the traffic destined to d for conciseness and suppose that congestion is detected on the next road segment of intersection Y. Although the infrastructure with RSUs is shown here, another type, such as cell-phone-based infrastructure, is also applicable. FCR monitors traffic and knows the volume of flow destined to d from both intersections X and Y. FCR also measures the estimated travel time from X and Y to d . Although Y is closer to the congestion, rerouting vehicles from X is better for both global optimization and users' benefit since the travel time loss is smaller. If the reroutable traffic volume at X is insufficient to cope with the congestion, FCR reroutes traffic at both X and Y to satisfy the required volume to divert the congested road segment.

B. System Model

A directed and weighted graph $G = (I, R)$ represents the road network, where I is the set of intersections and R is the

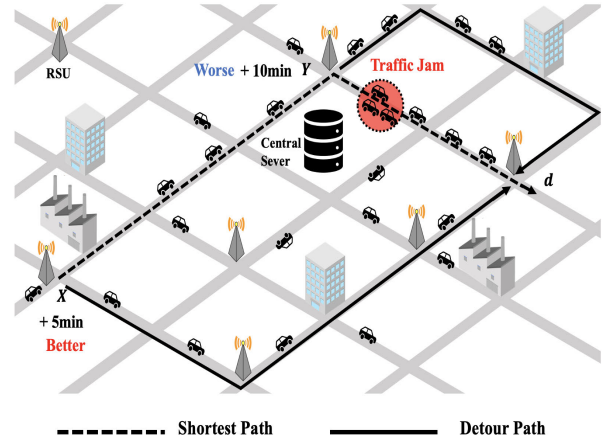


Fig. 1. Overview of our method.

set of road segments. A weight function for every road segment is defined as $w(\cdot)$, where $w(r)$ for road segment $r \in R$ is a positive value. Every intersection in the target map can be a source or a destination of a vehicle. For a source $s \in I$ and a destination $d \in I$, $P_{(s,d)}$ represents the shortest path from s to d under the weight function $w(\cdot)$. We also let $P_{(s,d)}$ be a set of road segments in the shortest path for conciseness. FCR calculates detour paths based on the k -shortest paths; $B_{(s,d)}^i$ represents the i -th ($i \leq k$) detour path from s to d on $G' = (I, R - P_{(s,d)})$ under the weight function $w(\cdot)$. Namely, the k detour paths are all edge-disjoint with the shortest path $P_{(s,d)}$ so that they all fork at the source intersection s . For a path P , $w(P)$ is the weight of P defined as the sum of the weights of all road segments included in P , i.e., $w(P) = \sum_{r \in P} w(r)$. The difference in weight between the shortest path and the k -th detour path from s to d is designated by $C_{(s,d)}^k = w(B_{(s,d)}^k) - w(P_{(s,d)})$. V is a set of vehicles in the road network.

Let V_{len}^{avg} be the average vehicle length and g_{min} be the minimum distance between two vehicles. For a road segment $r \in R$, let len_r be the segment length, $lane_r$ be the number of lanes, and n_r^t be the number of vehicles on road r at time t . The vehicle density of r at time t is expressed as follows,

$$K_r^t = \frac{n_r^t}{len_r \times lane_r}. \quad (1)$$

The maximum number of vehicles that can exist on r is $n_r^{max} = \frac{len_r}{V_{len}^{avg} + g_{min}} \times lane_r$. The maximum vehicle density of r is given as

$$K_r^{max} = \frac{n_r^{max}}{len_r \times lane_r}. \quad (2)$$

If the ratio of vehicle density to the maximum vehicle density of road r exceeds a predefined threshold δ at time t , i.e., $\frac{K_r^t}{K_r^{max}} \geq \delta$, r is regarded as a congested road. Accordingly, we detect congestion on a road segment based on vehicle density.

The traffic volume of a road segment $r \in R$ at time t is defined as $F_r^t = v_r^t K_r^t$ where v_r^t is the average vehicle speed of r at t and K_r^t is the average vehicle density of r at t . To define the capacity of $r \in R$, we use the Greenshield's model [40]. The model indicates that there is a linear relationship between

the vehicle speed and traffic density of a road segment $r \in R$, i.e., $\frac{v_r^t}{v_r^{max}} + \frac{K_r^t}{K_r^{max}} = 1$ where v_r^{max} is the given maximum speed of r . Let $v_r^* = pv_r^{max}$ be the vehicle speed that maximizes traffic volume of $r \in R$, then the corresponding density of r is expressed as $K_r^* = (1 - p)K_r^{max}$. Here, $p = \frac{1}{2}$, and the maximum traffic volume is $v_r^* K_r^* = \frac{1}{4} v_r^{max} K_r^{max}$. If we let t_i^g be the green light duration of the traffic light at the intersection $i \in I$ that the road segment r enters, and t_i^{cycle} be the period of traffic light at intersection i , the traffic capacity of the road segment r is defined as follows,

$$F_r^{cap} = v_r^* K_r^* \times \frac{t_i^g}{t_i^{cycle}}, \quad (3)$$

where $F_r^{cap} - F_r^t$ is defined as the residual capacity of r .

The traffic capacity of intersections differs depending on the direction of vehicle movement. In particular, the traffic capacity of right-turning traffic (in countries with left-hand traffic) tends to be much lower than that of straight-ahead or left-turning traffic because it depends on the traffic volume coming from the opposite direction. Thus, FCR considers the traffic capacity of intersections for each moving direction to estimate the capacity of detour paths more accurately. Let $i \in I$ be the intersection that $r \in R$ goes in, and $o(r) \in R$ be the opposite-direction road segment entering i . Then, $F_{o(r)}^t$ represents the traffic volume of $o(r)$. For the given road segments r_1 entering an intersection and r_2 outgoing the same intersection, the traffic capacity moving from r_1 to r_2 at time t is defined as follows.

$$F_{r_1 \rightarrow r_2}^{cap,t} = \begin{cases} F_{r_1}^{cap}, & \text{go straight or turn left,} \\ f(F_{o(r_1)}^t), & \text{turn right,} \end{cases} \quad (4)$$

where $f(\cdot)$ is the function that expresses the relation between the capacity and the traffic volume in the reverse direction. Although some papers in the literature studied the traffic at intersections on traffic light control [48], traffic capacity has not been well modeled yet. Therefore, in this paper, we define $f(\cdot)$ based on simulation, as shown in Section IV-B.

Assume that, at time t , the shortest path $P_{(s,d)}$ from intersection $s \in I$ to $d \in I$ passes through road segment r , and the congestion is detected on r . The traffic volume from s to d at time t is defined as the traffic volume going through $P_{(s,d)} = (r_1, r_2, \dots, r_n)$, which is written as $F_{(s,d)}^t = F_{r_1}^t$, i.e., it is represented by the road closest to s that has the latest volume value of the flow. On the other hand, if we guide vehicles in this flow from s to d to reroute to $B_{(s,d)}^k = (r_1, r_2, \dots, r_n)$, the maximum traffic volume that can be rerouted to $B_{(s,d)}^k$ is the same as the residual traffic capacity of $B_{(s,d)}^k$ defined as follows.

$$A_{(s,d),k}^t = \min(\min_{i \leq n-1} (F_{r_i \rightarrow r_{i+1}}^{cap,t} - F_{r_i \rightarrow r_{i+1}}^t), \min_{i \leq n} (F_{r_i}^{cap} - F_{r_i}^t)), \quad (5)$$

where $F_{r_i \rightarrow r_{i+1}}^t$ is the traffic volume that moves from r_i to r_{i+1} at time t .

The notations defined above are shown in the Table I.

An example of the reroutable traffic volume is shown in Fig.2. We focus on the flow from intersection s to d ,

TABLE I
NOTATION

Symbol	Description
G	Road network
I	Intersection set
R	Road segment set
G_M	Overlay network to compute detour paths
I_M	Intersection set in overlay network
R_M	Road segment set in overlay network
$w(r)$	Weight of road segment $r \in R$
$w(P)$	weight of path P
$P_{(s,d)}$	The shortest path from intersection $s \in S$ to $d \in D$
$B_{(s,d)}^k$	k -th detour path from intersection $s \in S$ to $d \in D$
$C_{(s,d)}^k$	The difference of weight between the shortest path and detour path $B_{(s,d)}^k$ from s to d
V	Vehicle set
V_{len}^{avg}	Average of vehicle length
g_{min}	Minimum inter-vehicle distance
K_r^t	Vehicle density of r at time t
n_r^t	Vehicle number on r at time t
v_r^t	Average speed of the vehicles on r at time t
K_r^{max}	Maximum vehicle density of r
n_r^{max}	Maximum vehicle number of r
len_r	Road length of r
$lane_r$	The number of Lanes of r
δ	Threshold for congestion detection
t_i^g	Green light duration of traffic light on intersection $i \in I$
t_i^{cycle}	The period of traffic light on intersection i
F_r^t	Traffic volume of r at time t
$F_{(s,d)}^t$	Traffic volume of flow (s, d) at time t
$F_{r_1 \rightarrow r_2}^t$	Traffic volume moving from r_1 to r_2
F_r^{cap}	Traffic capacity of r
$F_{r(o)}^t$	Traffic volume of the oncoming road $r(o)$
$F_{r_1 \rightarrow r_2}^{cap,t}$	Traffic capacity from r_1 to r_2
$A_{(s,d),k}^t$	Reroutable traffic volume for $B_{(s,d)}^k$ at time t

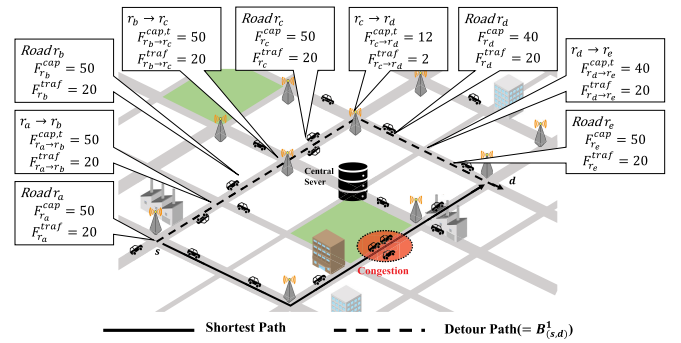


Fig. 2. Calculating reroutable traffic volume of detour path.

and assume that congestion is detected on the shortest path. A detour path $B_{(s,d)}^1$ is shown as a broken line, and we compute the traffic volume reroutable to the detour path. The capacity and the traffic volume of each road segment and intersection are shown in the figure, where the traffic volume is denoted by F_r^{traf} instead of F_r^t . From those values, we obtain the volume is $A_{(s,d)}^t = 10$ according to Equation (5), which is the same as $F_{r_c \rightarrow r_d}^{cap,t} - F_{r_c \rightarrow r_d}^{traf}$.

C. Traffic Monitoring

FCR constantly monitors traffic of the entire target area and knows the volume of reroutable traffic computed with Equation (5) for each traffic flow corresponding to any intersection pair (s, d) . Although the vehicle traffic is inherently

dynamic, this flow-based monitoring can follow traffic dynamics because transitions in traffic conditions are continuous and predictable on a 10-minute or hourly scale [49], [50]. However, it is essential to note that if we reroute vehicles to deal with congestion, the traffic transitions are no longer continuous because of the rerouted traffic flow. This causes a significant error in estimating the reroutable traffic volume calculated with Equation (5). To avoid this problem, FCR defines the *base traffic volume* for each road segment as the traffic volume before congestion occurs, which is used to estimate the reroutable traffic volume of the detour paths. The *base traffic volume* for each road segment is determined when congestion is detected, and held that value until all congestion subsides. Specifically, when we compute the reroutable traffic volume, we use the modified version of Equation (5), in which the base traffic volume is used in place of the traffic volume at time t as follows.

$$A_{(s,d),k}^t = \min(\min_{i \leq n-1} (F_{r_i \rightarrow r_{i+1}}^{cap,t} - F_{r_i \rightarrow r_{i+1}}^{traf}), \min_{i \leq n} (F_{r_i}^{cap} - F_{r_i}^{traf})), \quad (6)$$

where $F_{r_i}^{traf}$ and $F_{r_i \rightarrow r_{i+1}}^{traf}$ are the base values of traffic volume measured before congestion occurs. Namely, in FCR, we set the traffic volume $F_{r_i}^{traf}$ in place of $F_{r_i}^t$ when congestion is detected, and use this base value until all congestions are eliminated. Note that this implicitly assumes that traffic congestion will be resolved in a relatively short period of time. Note also that the residual traffic capacity $A_{(s,d),k}^t$ may be shared by multiple detour flows. The method for calculating the traffic volume diverted into each detour path is described in Sec. III-E.

To reduce the computational load, FCR specifies *major intersections* $I_M \subseteq I$ and computes the detour paths only for flows between major intersections (s, d) where $s, d \in I_M$. FCR structures an overlay network $G_M = (I_M, R_M)$ over G , and the detour paths are computed as the paths on G_M . This is practical because, in most cities, there are many narrow streets that are not suitable for detours. Since the detour paths are limited on G_M , we must choose a set of major intersections and their adjacencies such that G_M covers all possible detour paths and the paths between any adjacent major intersections $i_M \in I_M$ are uniquely determined on G .

FCR constantly monitors the traffic volume of each flow (s, d) where $s, d \in I_M$. Every time congestion is detected, the central server computes the detour paths and the reroutable traffic volume for each flow. FCR also monitors the average travel time for each flow, i.e., the average travel time to reach $d \in I_M$ from $s \in I_M$. FCR has the average travel time for each road and uses it as the weight of roads $w(\cdot)$ to compute the detour paths. Additionally, FCR monitors the number and speed of vehicles on each road segment for congestion detection and capacity calculations, as is already mentioned.

D. Rerouting Table

FCR builds a *rerouting table* when congestion on a road segment is detected, and the rerouting table contains the information for route guidance. When congestion expands and

multiple road segments are congested, the rerouting table is rebuilt for each congested road segment.

An example of the rerouting table is shown in Table II, where $L_{(s,d)}^k$ is the priority of $B_{(s,d)}^k$, $C_{(s,d)}^k$ is the estimated travel time loss of $B_{(s,d)}^k$ defined as $w(B_{(s,d)}^k) - w(P_{(s,d)})$, $E_{(s,d)}^k$ is the reroutable traffic volume of $B_{(s,d)}^k$, and $X_{(s,d)}^k$ is the cumulative value of $E_{(s,d)}^k$ of the detour paths with higher or equal priority in this table, i.e., $X_{(s,d)}^k$ is the sum of $E_{(s,d)}^k$ in this table that has higher or equal priority. The specific calculation of $E_{(s,d)}^k$ is described in Sec. III-E. Roughly, FCR builds the rerouting table by computing k -shortest paths for all flows (s, d) in G_M , sorting them using the travel time loss, and giving priorities accordingly. Namely, the rerouting table represents the list of rerouting paths used in the guidance, with their priority and capacity. Note that FCR builds a rerouting table for each congested road, and each entry of the rerouting tables has a unique priority number $L_{(s,d)}^k$ among all the rerouting tables.

When congestion on a road segment r is detected at time t , we reduce the traffic volume injecting to road r to mitigate the traffic congestion. We define the traffic volume to mitigate congestion as follows,

$$F_r^{exc} = F_r^{cap} - F_r^t. \quad (7)$$

Note that rerouting a larger volume of traffic will increase the total travel time of vehicles, while will reduce the time for congestion mitigation. Considering this trade-off, FCR injects less traffic into the congested road segment r than can be handled by r and reroutes the remaining traffic to a detour. Since F_r^t is the current volume that r can handle, even if the current injection volume to r is as large as F_r^{cap} , bypassing F_r^{exc} will reduce the volume injected to r to less than F_r^t , reducing the congestion on r .

We consider that the congestion will be resolved by guiding the volume βF_r^{exc} of vehicles that will pass the road r to the detour path, where $\beta \geq 1$ is the congestion expansion factor introduced to account for the growth of congestion in the near future. To mitigate the congestion on r , FCR determines the detour paths to apply in route guidance by selecting a set of rows in the rerouting table based on the priority that satisfies the following condition.

$$\alpha X_{(s,d)}^k \geq \beta F_r^{exc}, \quad (8)$$

where α is the estimated probability that the guided vehicles follow the route guidance.

If the cumulative volume of rerouted traffic (i.e., $\alpha X_{(s,d)}^k$) is larger than the traffic volume required to be diverted from the congested road (i.e., βF_r^{exc}), the congestion will be eliminated. Thus, FCR computes the minimum number $i (= L_{(s,d)}^k)$ that satisfies Equation (8), and applies route guidance with all rerouting paths having the priority less than or equal to i .

For instance, assume $F_r^{exc} = 85$, $\alpha = 0.7$ and $\beta = 1.3$. If we apply route guidance up to the sixth line in Table II (i.e., $i = 15$), $X_{(s,d)}^k = 165$. Thus, $\alpha X_{(s,d)}^k = 115.5$ vehicles per unit time will change their paths to avoid the congested road r . Since the traffic volume that will avoid r exceeds $\beta F_r^{exc} = 110.5$, the congestion will be eliminated by the route guidance with these six intersection pairs (s, d) .

TABLE II
REROUTING TABLE

priority $L_{(s,d)}^k$	source intersection $s \in S$	destination intersection $d \in D$	detour path $B_{(s,d)}^k$	arrival delay $C_{(s,d)}^k [\text{min}]$	traffic volume guided $E_{(s,d)}^k [\text{l/min}]$	cumulative $X_{(s,d)}^k [\text{l/min}]$
1	s_1	d_1	$B_{(s_1,d_1)}^1$	3	26	26
3	s_2	d_2	$B_{(s_2,d_2)}^1$	3	28	54
6	s_1	d_2	$B_{(s_1,d_2)}^1$	6	25	79
9	s_3	d_1	$B_{(s_3,d_1)}^1$	8	21	100
11	s_1	d_3	$B_{(s_1,d_3)}^1$	8	34	134
15	s_4	d_2	$B_{(s_4,d_2)}^1$	9	31	165
17	s_3	d_4	$B_{(s_3,d_4)}^1$	11	14	179
:	:	:	:	:	:	:
:	:	:	:	:	:	:
$i-1$	s_1	d_1	$B_{(s_1,d_1)}^3$	31	$E_{(s_1,d_1)}^3$	$X_{(s_1,d_1)}^3$
i	s_{14}	d_5	$B_{(s_{14},d_5)}^2$	33	$E_{(s_{14},d_5)}^2$	$X_{(s_{14},d_5)}^2$
:	:	:	:	:	:	:

Algorithm 1 Building Rerouting Tables

```

1: foreach  $r \in J$  do
2:    $T_r \leftarrow \emptyset$  ▷ initializing rerouting table
3:   compute  $F_r^{exc}$  ▷ traffic volume to be rerouted
4:   compute  $\mathcal{B}_r$  ▷ detour paths candidates
5:   sort  $\mathcal{B}_r$  ▷ sort by travel time loss
6: end foreach
7: while  $r \in J$  such that  $F_r^{exc} > 0$  exists do
8:    $r \leftarrow \arg \max_{r \in J} F_r^{exc}$ 
9:    $B_{(s,d)}^k \leftarrow \text{retrieveMin}(\mathcal{B}_r)$ 
10:  compute  $A_{(s,d),k}^t$ 
11:  compute  $H_{(s,d)}^{traf}$ 
12:   $E_{(s,d)}^k \leftarrow \min(F_{(s,d)}^{traf}, A_{(s,d),k}^t, H_{(s,d)}^{traf})$ 
13:  if  $E_{(s,d)}^k = 0$  then
14:    continue
15:  end if
16:   $F_r^{exc} \leftarrow F_r^{exc} - E_{(s,d)}^k$ 
17:  foreach  $r' \in B_{(s,d)}^k$  do
18:     $F_{r'}^{traf} \leftarrow F_{r'}^{traf} + E_{(s,d)}^k$ 
19:  end foreach
20:  foreach  $(r_1, r_2) \in B_{(s,d)}^k$  do
21:     $F_{r_1 \rightarrow r_2}^{traf} \leftarrow F_{r_1 \rightarrow r_2}^{traf} + E_{(s,d)}^k$ 
22:  end foreach
23:   $T_r \leftarrow T_r \cup \{B_{(s,d)}^k\}$ 
24: end while

```

E. Algorithm for Building Rerouting Tables

The algorithm to build rerouting tables when congestion on a set of road segments J is detected, is shown in Algorithm 1. In the algorithm, FCR computes the traffic volume F_r^{exc} that should be rerouted, for each congested road segment $r \in J$. FCR then computes detour paths and their estimated travel time, and inserts them into the rerouting tables, prioritizing the loss of travel time, until the required traffic volume F_r^{exc} is fulfilled. Every time FCR inserts a detour path, FCR updates the residual capacity of the road segments in the detour path. By repeating this, FCR finally completes the rerouting tables for all $r \in J$.

In lines 1-6, FCR makes variable initialization. For each congested road segment $r \in J$, the traffic volume that should

be rerouted, i.e., F_r^{exc} , is computed according to Equation (7). FCR also obtains a set of detour paths for $r \in J$ as $\mathcal{B}_r$. The detour paths are computed based on travel time given as the weight function $w(\cdot)$. Specifically, FCR identifies a set of flows (s, d) where $s, d \in I_M$ and $r \in P(s, d)$, and computes a set of k -shortest paths on $G'_M = (I_M, R_M - P(s, d))$ for the flows (s, d) , obtaining a set of detour paths $\mathcal{B}_r$. FCR then sorts the detour paths in $\mathcal{B}_r$ by the travel time loss, which is the difference between the travel time of the shortest path and the detour path.

In the while loop in lines 7-24, FCR inserts detour paths one by one to the rerouting tables. In lines 7-9, it first selects the congested road segment $r \in J$ with the maximum F_r^{exc} . It next retrieves the detour path $B_{(s,d)}^k$ with the minimum travel time loss from $\mathcal{B}_r$, which is the path to insert the rerouting table T_r . In line 10, FCR computes its reroutable traffic volume $A_{(s,d),k}^t$ according to Equation (6).

In line 11, FCR computes the *reserved traffic volume* $H_{(s,d)}^{traf}$, a constraint that accounts for the traffic volume already planned to be rerouted at the downstream intersections of s in flow (s, d) . If the detour path $B_{(s',d)}^k$ such that $s' \in P(s, d)$ has already been added to $\mathcal{B}_r$ for any $r \in J$, FCR must guarantee the traffic volume $E_{(s',d)}^k$ to enter s' . Here, the traffic volume of flow (s, d) entering s' is written as $F_{(s',d)}^{traf}$, and the traffic volume to reserve for s' is written as $\sum_{(s',d,k') \in \mathcal{B}_r, r \in J, 1 \leq k' \leq K} E_{(s',d)}^{k'}$, where K is the number of shortest paths for each pair (s, d) computed by the k -shortest-path algorithm. Therefore, the maximum traffic volume for rerouting at s is written as

$$H_{(s,d)}^{traf} = \min_{s' \in P(s,d)} (F_{(s',d)}^{traf} - \sum_{(s',d,k') \in \mathcal{B}_r, r \in J, 1 \leq k' \leq K} E_{(s',d)}^{k'}), \quad (9)$$

where $\mathcal{B}_r$ is the set of rerouting table entries for $r \in J$ at that time during the execution of Algorithm 1. In line 12, FCR determines the traffic volume to reroute on $B_{(s,d)}^k$, which is represented by $E_{(s,d)}^k$, as the smallest value among the flow's traffic volume $F_{(s,d)}^{traf}$, the detour-path capacity $A_{(s,d),k}^t$, and the reserved traffic volume $H_{(s,d)}^{traf}$.

In lines 13-23, FCR updates the capacity of road segments in the detour path $B_{(s,d)}^k$ and inserts $B_{(s,d)}^k$ into the rerouting table $\mathcal{B}_r$. Lines 13-15 skip processing $B_{(s,d)}^k$ if its capacity is zero.

Line 16 subtracts the rerouting volume from F_r^{exc} . Lines 17-19 add the rerouting volume to the base traffic volume of the road segments in $B_{(s,d)}^k$. Lines 20-22 similarly add the rerouting volume to the base traffic volume of intersections in $B_{(s,d)}^k$. Finally, in line 23, FCR inserts $B_{(s,d)}^k$ into the corresponding rerouting table $\mathcal{B}_r$ where the priority number is incremented each time an entry is inserted.

Note that we share the residual capacity of roads with multiple detour paths by temporarily updating the base traffic volume of each road and intersection. When the detour volume for $B_{(s,d)}^k$, i.e., $E_{(s,d)}^k$, is determined, the volume is added to the base traffic volume in lines 18 and 21 of Algorithm 1, with which we calculate the residual capacity $A_{(s,d),k}^i$ for the next detour path $B_{(s,d)}^k$.

F. Route Guidance Behavior

Once congestion is detected for a set of road segments J , FCR fixes the base traffic volume for each road segment and builds rerouting tables for each $r \in J$. The rerouting tables are rebuilt at regular time intervals or when a set of congested road segments changes to follow road traffic dynamics.

The rerouting table is a set of guidance instructions for flows, and the route guidance operation of FCR is performed in the following steps. First, (1) FCR checks every vehicle that approaches every intersection $s \in I_M$ and identifies the flow to which each vehicle belongs. Next, (2) FCR checks if the vehicle meets entries of the rerouting tables and guides to reroute it if applicable. This is done for every vehicle whenever it approaches an intersection in I_M .

In step (1), to identify the flow (s, d) to which a vehicle (at intersection s) belongs, FCR obtains the destination from the vehicle. If the destination is outside of the target area (i.e., outside of the map), the intersection at the boundary of the target area on the shortest path to the destination is identified as the corresponding intersection d . We can implement this process by mapping destinations and their corresponding intersections d .

In step (2), if vehicles in a flow match an entry of detour path $B_{(s,d)}^k$, FCR chooses vehicles in the flow probabilistically such that the volume of the chosen vehicles is $E_{(s,d)}^k$, and guides them to reroute. (Specifically, in our simulation, the appropriate probabilities are calculated from the latest flow measurements, and the vehicles are guided accordingly.) If a flow matches multiple entries in the set of rerouting tables, FCR does the same. Specifically, FCR, for each entry, probabilistically chooses the corresponding volumes of vehicles and guides the vehicles according to the assigned entries.

IV. EVALUATION

A. Method

We performed a simulation to evaluate the performance of the proposed method FCR. We used SUMO (Simulation for Urban MObility) [41] as the traffic simulator and TraCI (Traffic Control Interface) [44] as the interface for traffic control. We obtained the road map of Osaka city, the second-largest city in Japan, from OpenStreemap [42]. We defined an overlay network G_M on the city road map G by carefully

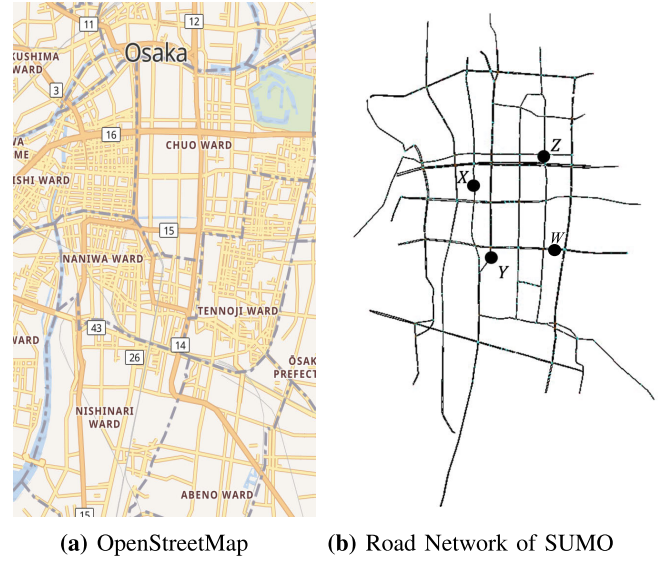


Fig. 3. Road network of Osaka city.

choosing major intersections and roads among them such that all major roads that are preferably used for detour paths are included in G_M . As a result, the number of major intersections $|I_M|$ is 77. Traffic signals are installed at all the applicable intersections with a cycle time of 120 seconds. We generate traffic flows between all pairs of boundary intersections such that the traffic volume on each road fits an official survey [43] of Japan. We create vehicles at regular time intervals for each flow according to the flow volume assigned to the flow. In Fig. 3(a), we show the Osaka city map, and in Fig. 3(b), we show the overlay network I_G built on it.

We aim to test the traffic efficiency in a heavy-load scenario where several major roads are simultaneously closed to compare the ability of several route guidance methods. We ran a simulation 30000 seconds long, and at 1800 seconds after beginning the simulation, we invoked traffic congestion by closing the roads X, Y, and Z in Fig. 3(b). Roads X, Y, and Z are chosen as the roads where the traffic load is particularly high on the map. We compared the performance of the proposed method FCR with major route guidance strategies in the centralized approach, that is, DSP [10], AR*, RkSP, EBkSP [11], and PkSP [14]. The main criteria for evaluation are the average travel time and the total congestion time.

In the simulations, we used several default settings in SUMO where the vehicle length is 5 m, the minimum inter-vehicular distance is 2.5 m, and the vehicle following model is Krauss [45]. The time interval for updating rerouting tables in a congestion state is 300 seconds, and the threshold δ for determining whether a road is congested is 0.5. The ratio of drivers following route guidance is 0.7, and the estimated ratio of drivers following route guidance is $\alpha = 0.8$. The traffic congestion expansion factor is $\beta = 1.3$. We summarize the simulation settings in Table III.

B. Preliminary Experiment: Right-Turning Capacity

We introduced a function $f(\cdot)$ to estimate right-turn capacity (or left-turn capacity in countries with right-hand traffic) in

TABLE III
SIMULATION SETTING

Item	Values
Simulator	SUMO
Target area	Osaka city
Number of intersections	7126
Number of road segments	25040
Number of major intersections I_M	77
Number of boundary road segments	26
Traffic signal cycle	120s
Maximum speed of vehicle	60km/h
Vehicle following model	Krauss
Vehicle length	5m
Minimum inter-vehicular distance	2.5m
Total number of vehicles	30000
Time interval for detecting congestion	300s
Number of closed road	3
δ	0.5
Compliance rate	0.7
α	0.8
β	1.3
k	3
L	2

Equation (4) in Section III-B. In this subsection, we provide a definition of $f(\cdot)$ based on simulation results.

We prepared a road network as shown in Fig. 4 to test the road capacity with various traffic volumes. There are three intersections, and each road has two lanes for each direction. Vehicles from the left go straight to the right with various volumes. We measure the traffic volume which comes from the right and turns right at the central intersection to go upward. Other traffic parameters, such as the traffic light cycle, are the same as the main simulation, as shown in Table III.

Figure 5 shows the relationship between the traffic volume from the left (seen on the horizontal axis) and the right-turning capacity (seen on the vertical axis). We approximated it by a quadratic equation and defined the function as $f(x) = 0.0157x^2 - 1.6886x + 85.275$ where x is the reverse-direction traffic volume. Although right-turning capacity may differ depending on intersection structure or parameters such as the number of lanes, we apply this definition to all intersections for simplicity. Since all detour paths are defined on major intersections and roads, this definition would have some validity.

C. Preliminary Experiment: Best Parameter Values

We determine the parameter values of k and L for comparison methods through simulation. We find the values best for comparison methods through simulation for a fair evaluation and use them in the main simulation. Here, k is the number of paths computed in the k -shortest paths computation, and L is the maximum distance of vehicles from the congested road to provide route guidance. That is, if $L = 1$, only vehicles on the road neighboring the congested road are guided for the detour paths in all the conventional methods in the simulation. The simulation parameters are the same as the main simulation shown in Table III.

Figure 6(a) shows the results with $k = 3$ and various L . From the result, $L = 2$ is the best for most conventional methods. A small value of L results in the insufficient ability of congestion mitigation, while a large value will increase the number of redundant detouring vehicles. Since $L = 2$ is

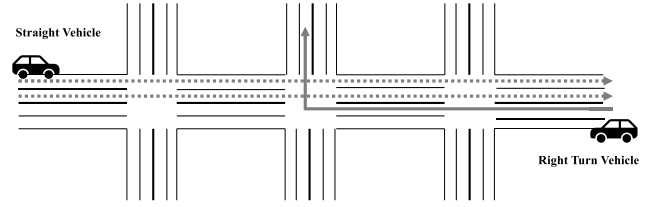


Fig. 4. Road network for calculating traffic capacity of right turn.

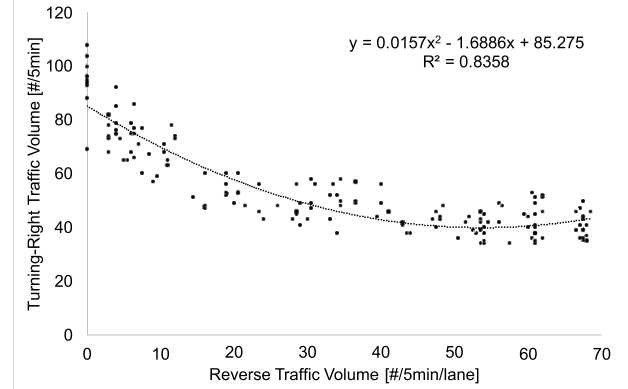


Fig. 5. Relationship between traffic volume on oncoming road and traffic capacity for right turn.

the best in our simulation scenario, we use this value in the main simulation. Fig. 6(b) shows the results with $L = 2$ and various k , which shows that $k = 3$ is the best value, with the trade-off where a small k reduces load balancing performance while a large k results in low-quality redundant routes used by vehicles. From the result, we adopt $k = 3$ in the main simulation for all the k -shortest-paths-based methods RkSP, EBkSP, PkSP, and FCR.

D. Main Result: Global Performance

Figure 7(a) shows the average travel time of all vehicles to reach their destinations in the main simulation. Here, the *Baseline* means the case where no road is closed so that no congestion occurs, and the *No Rerouting* means the case where roads are closed without route guidance. We see that all route guidance methods significantly reduce travel time, and especially, FCR marks the minimum travel time among all of the methods. Specifically, FCR improves average travel time by about 43% while DSP, RkSP, EBkSP, PkSP, and AR* do by about 18%, 15%, 27%, 33%, and 33%, respectively.

Figure 7(b) shows the total congestion time in the simulation. We also see that FCR eliminated traffic congestion the fastest even though FCR reroutes vehicles farther from the congested roads than the conventional methods.

Figure 7(c) shows the number of rerouted vehicles in each method. This result indicates that the number of rerouted vehicles in FCR is significantly smaller than that of the other methods. Considering that the travel time and the congestion time of FCR are shorter than the other methods (as shown in Figs. 7(a) and 7(b)), this means that the contribution of detouring per vehicle in FCR to the global optimization is significantly larger than the other methods.

Figure 7(d) shows the CO_2 emissions of each method calculated with an emission model [46]. Specifically, in the

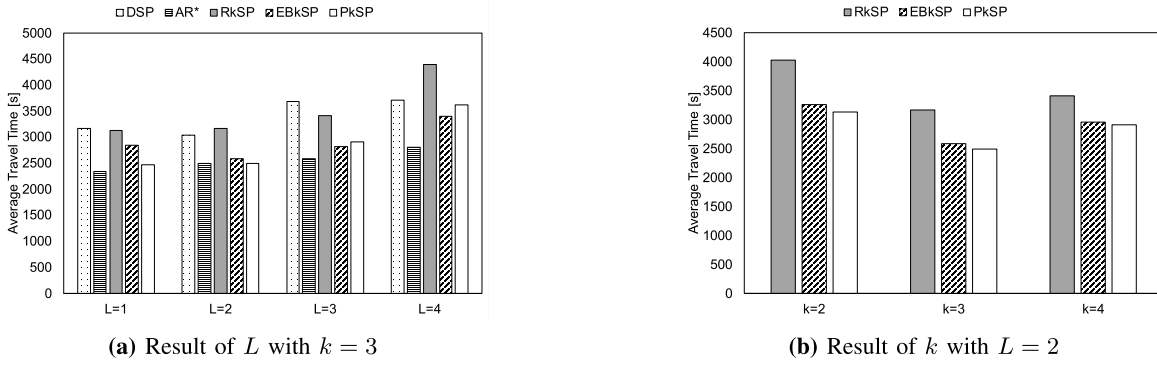


Fig. 6. Results of preliminary evaluation.

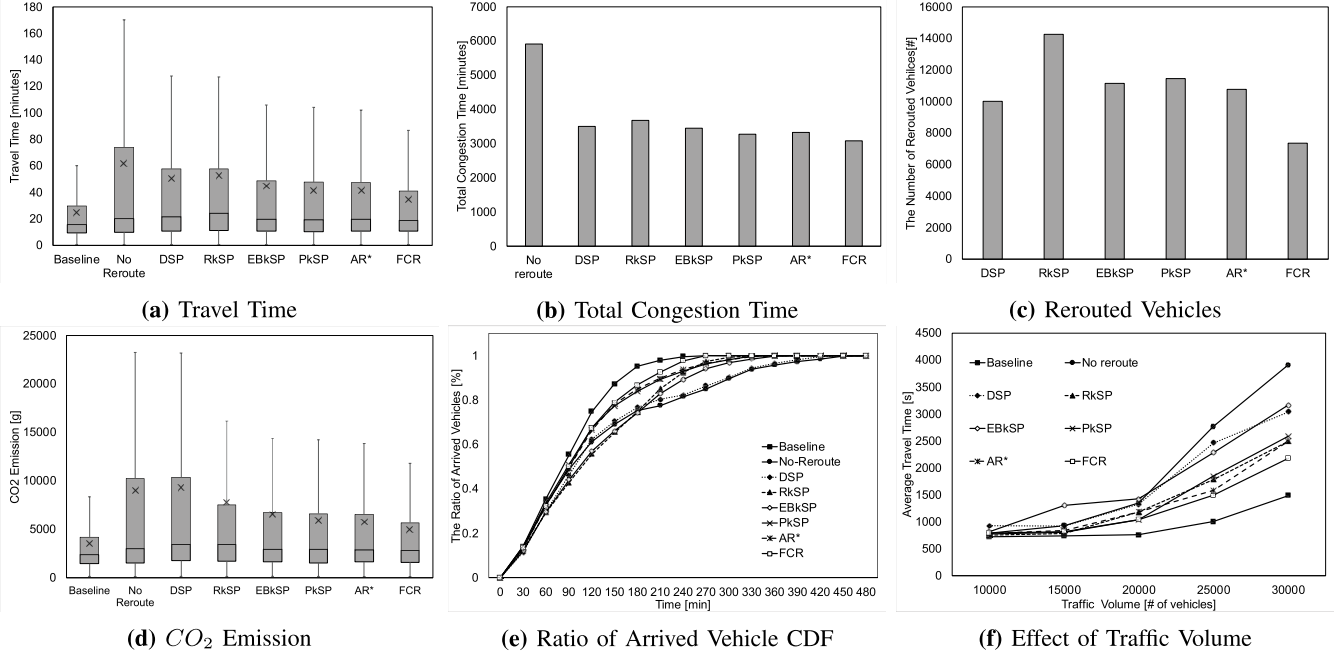


Fig. 7. Main result.

model, the emission of each vehicle is determined by the speed and acceleration of the vehicle and the emission factor given in HBEFA (Handbook Emission Factors for Road Transport) [47]. The figure also shows that FCR has superior performance compared to the others.

Figure 7(e) shows the cumulative distribution of arrived vehicles for each method. The CDF curve in FCR is the closest to the *baseline*, and particularly the arrival completion time is far better than others, which is about 17% better than AR* and 29% better than PkSP.

Figure 7(f) shows the average travel time with various traffic volumes, where we varied the traffic volume per flow. This result demonstrated that FCR stably provides good performance with any traffic volume.

Those results show that FCR is superior to other methods from the viewpoint of global optimization.

E. Main Result: User's Benefit

As we have noted previously, fairness and benefit of users are essential to consider the feasibility of the route guidance systems because users will not follow the system's guidance if they do not rely on the system.

Figure 8 compares the average travel time of the vehicles that follow the reroute guidance and those that do not. First, we see that the travel time of all categories is the smallest in FCR. We also find that, in FCR, the travel time of the vehicles that followed guidance is shorter than those that did not, meaning that vehicles will benefit if they follow the rerouting guidance. This is a favorable property for the actual application of the route guidance system.

Another finding from the result is that the travel time of all vehicles is shorter than the rerouted vehicle in FCR. The reason why this happened is as follows. First, FCR's rerouting paths are by nature longer than other methods because FCR invokes rerouting from the intersection farther from the congested roads. Second, the methods except FCR led to secondary congestion on their detour paths and increased the travel time of the vehicles that were not guided. From the user's viewpoint, although vehicles are sometimes guided to reroute at a farther location from congestion, the detour paths are lower loss, less congested than other methods, and lead to destinations with less time if they follow the guidance. In addition, as shown in Figs. 7(c) and 8, a relatively small number of vehicles reroutes to benefit all vehicles around

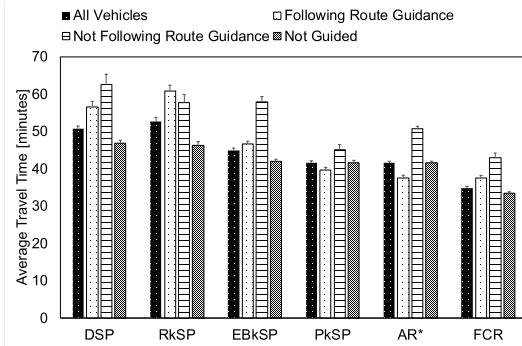


Fig. 8. Effect of rerouting.

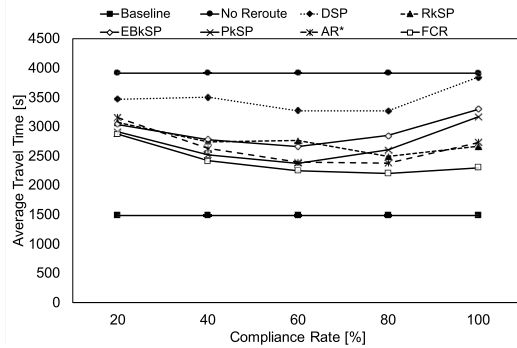


Fig. 9. Effect of compliance rate.

the map. As a result, in FCR, users have more benefits and satisfaction than other conventional methods and will gain motivation to follow the guidance.

F. Main Result: Effect of Road Closure

We further show the result on the effect of users' compliance rate. Fig. 9 shows the performance in which we changed the compliance rate of users. Specifically, we varied the rate of vehicles to follow the rerouting guidance between 20-100%. The result shows that the conventional methods worsen as the compliance rate approaches 100%. This is caused by the secondary congestion in the rerouting paths, which occurs due to insufficient coordination among the detour paths. Among all, FCR (with $\alpha = 0.8$) is the most stable and keeps the lowest travel time using capacity-aware detour-path coordination. This result indicates that FCR is the most robust against variation in the compliance rate of users.

Another simulation was performed to investigate the effect of road closures on detour routes. Specifically, four levels of road closure were tested. That is, the four combinations of roads X, XY, XYZ, and WXYZ shown in Fig. 3(b) were closed, where the traffic volumes without closure were X, Y, Z, and W in descending order. All other conditions are the same as in the above scenario except that the total number of vehicles is 20000.

Figure 10 shows the average vehicle travel time; FCR performs best when the number of closed roads is three or more, but the other methods sometimes outperform FCR when there are one or two closed roads. The reason for this is that, compared to FCR, which performs detour probabilistically, the conventional methods divert all vehicles traveling

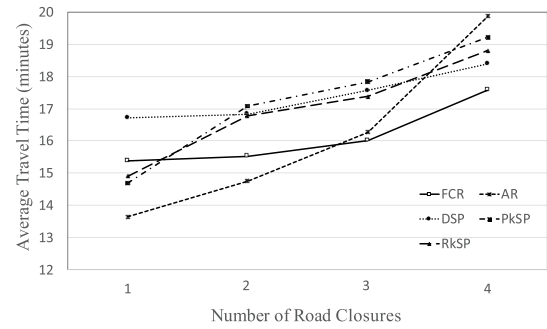


Fig. 10. Travel time (average).

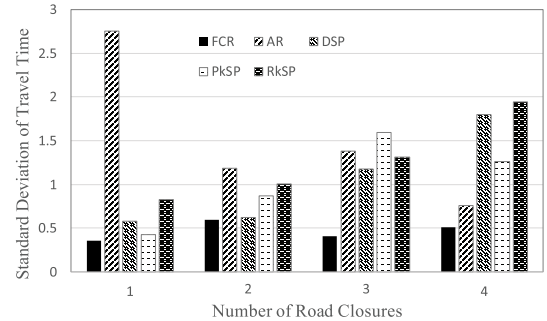


Fig. 11. Travel time (standard deviation).

near the congestion and are therefore more effective when traffic volume is low. However, when three or more roads are closed, the capacity of the detour roads becomes small, and the capacity-aware multipath property of FCR becomes effective. The conventional methods result in severe secondary congestion on the detour roads, which worsens the traffic situation.

Figure 11 shows the standard deviation of the average travel time over repeated simulations; FCR has the smallest standard deviation, indicating that FCR's performance is the most stable and reliable for users.

Figure 12 shows the congestion duration of all congested roads in the instances that have close travel time selected from the three-road-closure cases for each of FCR and AR*. We see that for both methods, congestion is detected on many roads. This is because congestion occurs as a chain of traffic jams; When a road is closed, some vehicles are forced to change their route, causing congestion on the new route, which causes further detours. The results in Fig. 12 show that the chain of congestion occurs in both scenarios. (Even in FCR, which is aware of the capacity of detour routes, congestion chains occur due to capacity estimation errors and capacity limitations of detour routes.) However, FCR has a shorter congestion time with a larger number of roads than AR*. This means that traffic volumes in FCR are more widely distributed across the road map than in AR*, and FCR adjusts to changing road conditions more quickly than AR*.

G. Computational Cost

FCR reduces the computational cost by (1) introducing flow-based rerouting path computation and by (2) limiting

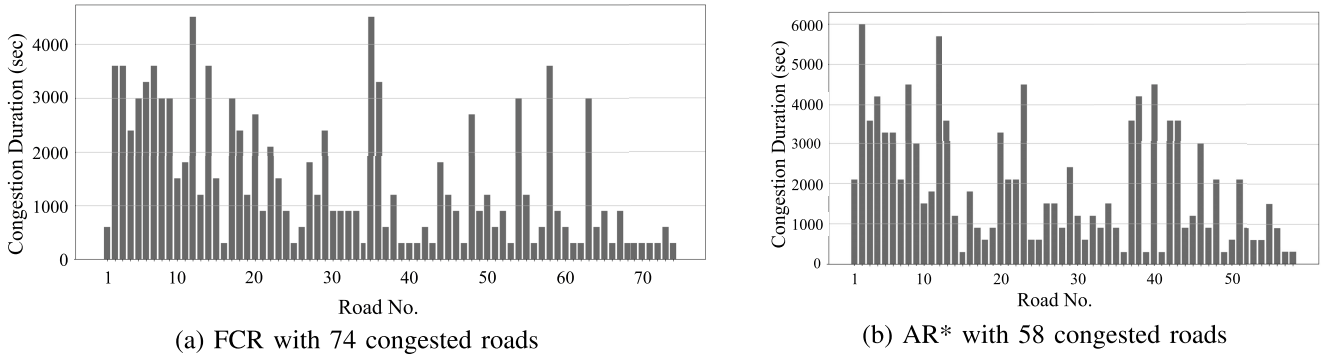


Fig. 12. Congestion duration.

detour paths on the sparse overlay network G_M . In (1), FCR measures traffic volume in the unit of traffic flow and computes rerouting paths per flow at regular intervals. In (2), by following the principle that the detour path should not use narrow streets of local life, FCR limits detour paths to major roads connected by major intersections. Since the number of nodes in the overlay network G_M is limited, the computational cost is significantly reduced. Although other centralized route guidance strategies can adopt these two mechanisms, past studies in the literature have regarded the centralized approach as significantly high cost mainly because they compute rerouting paths per vehicle [22], [27].

The computational load of FCR is mainly due to the computation of the detour path candidates B_r in line 4 of Algorithm 1. Here, we first compute the shortest path for every pair of intersections (s, d) where $s, d \in I_M$ to identify the flows that include congested road r . Next, we compute k -shortest paths for every identified pair of intersections. The time complexity of k -shortest paths computation is practically $O(kn(m + n \log n))$ [33], where n is the number of nodes and m is the number of edges, although there have been some improvements. Thus, reducing the number of nodes and edges in the network, i.e., $n = |I_M|$ and $m = |R_M|$, is essential to reduce the computational complexity. Additionally, flow-based rerouting paths management enables FCR to compute a set of rerouting paths once in a certain interval instead of the computation per vehicle, reducing the computational load.

In our simulation scenario, we used the map of Osaka City, which is the second-largest city in Japan. We carefully selected $|I_M| = 77$ major intersections to cover the detour-paths candidates from 7126 intersections. (Note that those 7126 intersections include very small intersections without a traffic light.)

Due to code constraints, our implementation actually computes the detour paths between roads, not the paths between intersections. Since the number of roads connected to the major intersections is 214 and the boundary roads of our map is 26, we computed k -shortest paths for $(214 + 26) \times 26 = 6240$ pairs of source and destination roads. As we mentioned, our algorithm consists of two folds. For the first part, the computational time for the shortest paths computation for all pairs of I_M in our simulation was 3.269 s with a CPU Ryzen5950 3.4GHz and 64GB memory without parallel computation (i.e., computation with a single core). This

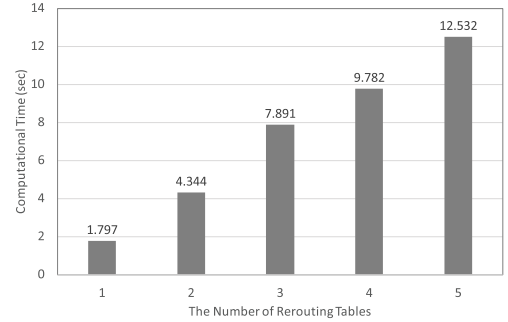


Fig. 13. Computational time for building rerouting tables.

computation identifies the flows to compute the k -shortest paths. For the second part, the computational time for building the rerouting tables via the k -shortest-paths computation is shown in Fig. 13, where we show the average time to build each number of rerouting tables. (We invoke the computation at regular intervals in the simulation, and different numbers of the rerouting tables are built each time.) The result shows that the computational load for building the rerouting table is sufficiently low. We additionally mention that the k -shortest paths computation can be parallelized in the cloud so that we can treat a larger number of major intersections in practice.

V. DISCUSSION

It would be worth discussing practical issues associated with implementing the centralized approach. Given that it takes 12 seconds to build five tables, as shown in Fig. 13, the impact of computational delay may need to be considered. However, as already noted, this results from a single non-parallelized consumer PC. A highly parallelized cloud server could easily reduce the latency to practical orders of magnitude.

Probabilistic vehicle selection for guidance can be fine-tuned in practice. Our implementation employs a simple probabilistic selection, in which probabilities are computed for each flow based on the latest traffic measurements, and we guide vehicles accordingly. However, practical performance may be improved by, for example, applying methods such as guaranteeing a certain volume of vehicles to be guided.

For a centralized management scheme to be effective in practice, penetration is critical for both traffic measurement and traffic optimization. Especially in the early stages of service, penetration is inevitably not sufficient. A similar

situation occurs when multiple companies compete to provide services. How to support traffic measurement with machine learning techniques such as [38] or how to achieve traffic optimization with a limited number of controllable vehicles can be a practical challenge.

Another issue is the range of guidance, i.e., the distance over which detours should be searched and guidance should be provided. Basically, the topology of the road network and the magnitude of congestion determine the appropriate range. In other words, the more congested the traffic, the greater the capacity of the detour route should be, and the capacity of the detour route depends on the topology of the road network. In fact, greater computational costs are required to consider a wider range of route guidance. Therefore, in practice, the range of route guidance should be considered for each specific case, depending on the road network and the magnitude of the expected congestion.

Finally, we set the rerouting table update interval to 300 seconds. This is probably one of the valid values, since traffic convergence after a rerouting table update takes at least a few minutes. A smaller interval would increase the computational load but may improve performance. The appropriate interval should be determined on a case-by-case basis.

VI. CONCLUSION

This paper proposes a new flow-based centralized route guidance architecture and a method called FCR to mitigate traffic congestion in urban areas. FCR measures traffic based on *flows* defined by pairs of intersections and computes rerouting paths on a flow-by-flow basis to reduce computational load to withstand actual system deployment. When congestion is detected, FCR computes the rerouting paths taking into account traffic capacity and achieves better global optimization performance worthy of centralized control. FCR selects rerouting paths based on the travel time loss, i.e., the time loss experienced by users by rerouting, which improves both the global performance and the users' satisfaction. Traffic simulation results showed that FCR achieves the best global performance among other centralized methods by rerouting fewer vehicles with a low computational load. It was also shown that FCR provides users benefits, indicating that they have good reason to follow the FCR's rerouting guidance. This suggests that a centralized architecture remains a viable option for the actual deployment of route guidance systems.

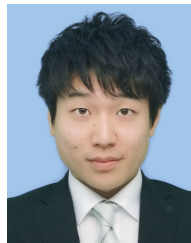
Several issues remain as future work. First, more accurate traffic capacity estimation based on rational traffic models is one of the key issues. In this paper, we used a simulation-based traffic capacity estimation of turning-right traffic in intersections, but in reality, the capacity varies depending on the situation. It is desirable to develop more accurate general capacity models that consider the patterns of traffic, traffic lights, etc. Second, a method to follow the dynamics of urban traffic even during congestion will be beneficial to improving performance. FCR uses a fixed traffic volume measured just before congestion occurs to estimate the capacity of the detour paths, which assumes that the congestion would be mitigated in a short time. However, traffic volume may fluctuate even

while the congestion continues. Following the dynamics in real-time would improve the performance.

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